

Short Answer Test:

- 1) a) Detects and recognize objects, ~~Analyze~~ and understand the scenes. Extracts useful information from an Images. 50
- b) face recognition in Smartphones and Security Systems.  
medical Image detection for disease Identification
- 2) a) feature Extraction is a process of Identifying important and detailed characteristics such as edges, corners, shapes, context etc..
- b) object recognition enables system to identify and classify the objects present in the Image, Improving Accuracy in applications like Surveillance, Diagnosis, and Autonomous driving.
- 3) a) low-level Image processing deals with basic operations such as Image Enhancement, noise removal, filtering, and contrast Improvement. The Input and output both are Images.

b)

mid-level

Extracts Image features  
Includes segmentation and feature Extraction

High-level

understand Image meaning  
Includes object recognition and scene interpretation

4) a) A pixel is the smallest unit of digital image brightness or color information.

b) Higher resolution means more pixels, resulting in better quality of image. Low resolution means less pixels resulting in poor quality of image.

5) a) Capture light from a scene and converts it into electrical signals that are processed into digital images.

b) Digital camera scanner.

6) a) Sampling is the process of dividing continuous image into discrete pixels.

b) Quantization converts continuous intensity values into a limited number of digital levels for computer storage and processing.

7) a) Image formation model explains how light from an object passes through a camera lens and forms an image on sensor.

a) Perspective projection is a method where three dimensional objects are projected onto a 2D image plane.

b) Illumination provides the light necessary to capture clear images. Proper lighting improves image quality and feature visibility.



Image processing enhances or transforms images, while computer vision interprets image content to understand.

a) Image processing is the technique of performing operation on images to improve quality.

| C.V (computer vision)                                                      | I.P (Image processing)                                |
|----------------------------------------------------------------------------|-------------------------------------------------------|
| understands image content<br>output is meaningful information or decisions | Enhances or modifies image<br>output is another image |

9) C.V is widely used in healthcare, transportation, agriculture, manufacturing etc...

a) C.V analyzes X-rays, MRI, CT scan images to detect diseases, helping doctors make accurate diagnoses.

b) C.V enables self-driving, traffic signals, pedestrian, vehicles and detects the surroundings.

10) Digital Images helps the computer vision to understand and analyze the scene and detect the motion.

a) pixels are smallest elements of digital image together they complete image by storing information of image

b) sampling determines the no. of pixels, while resolution